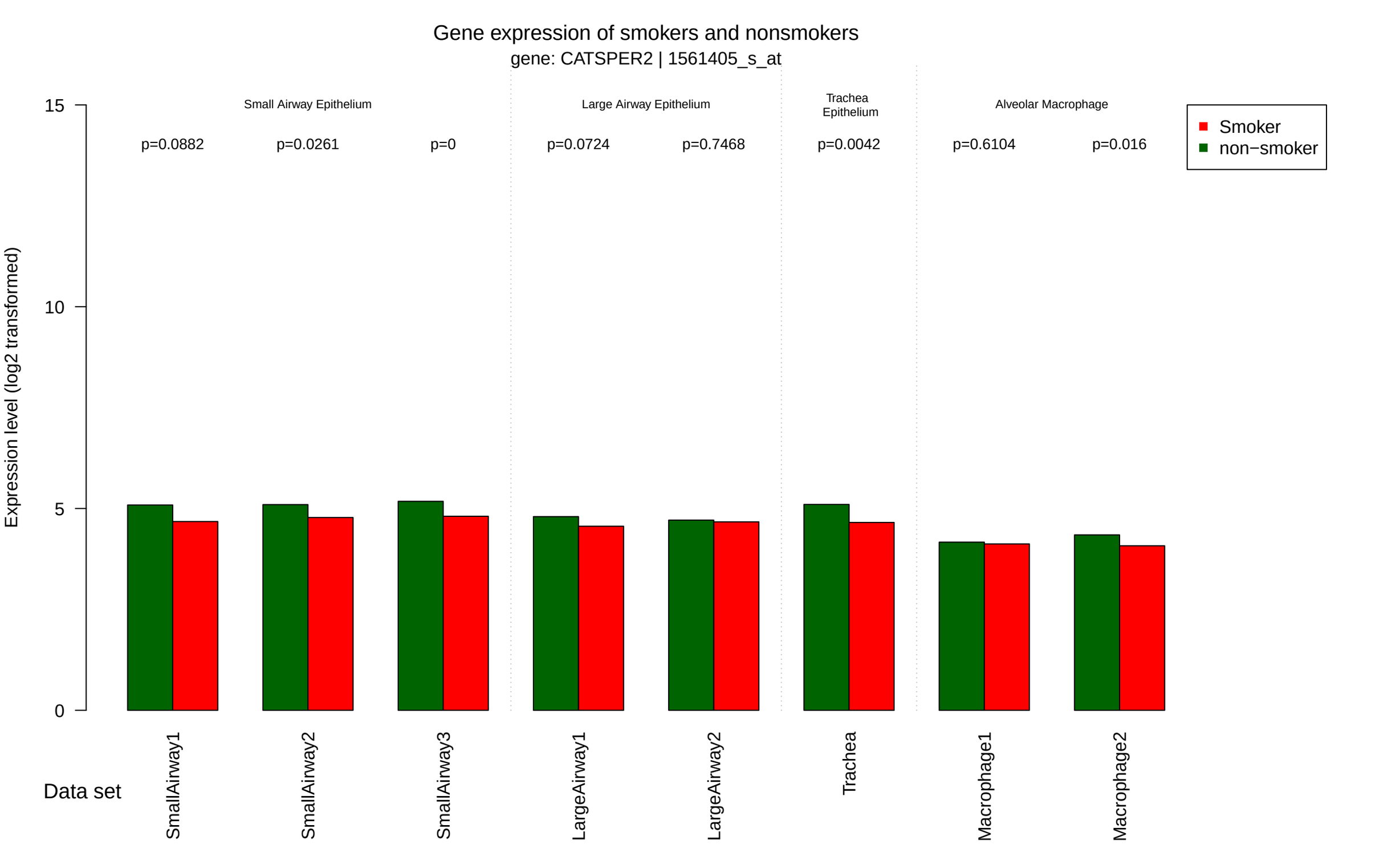
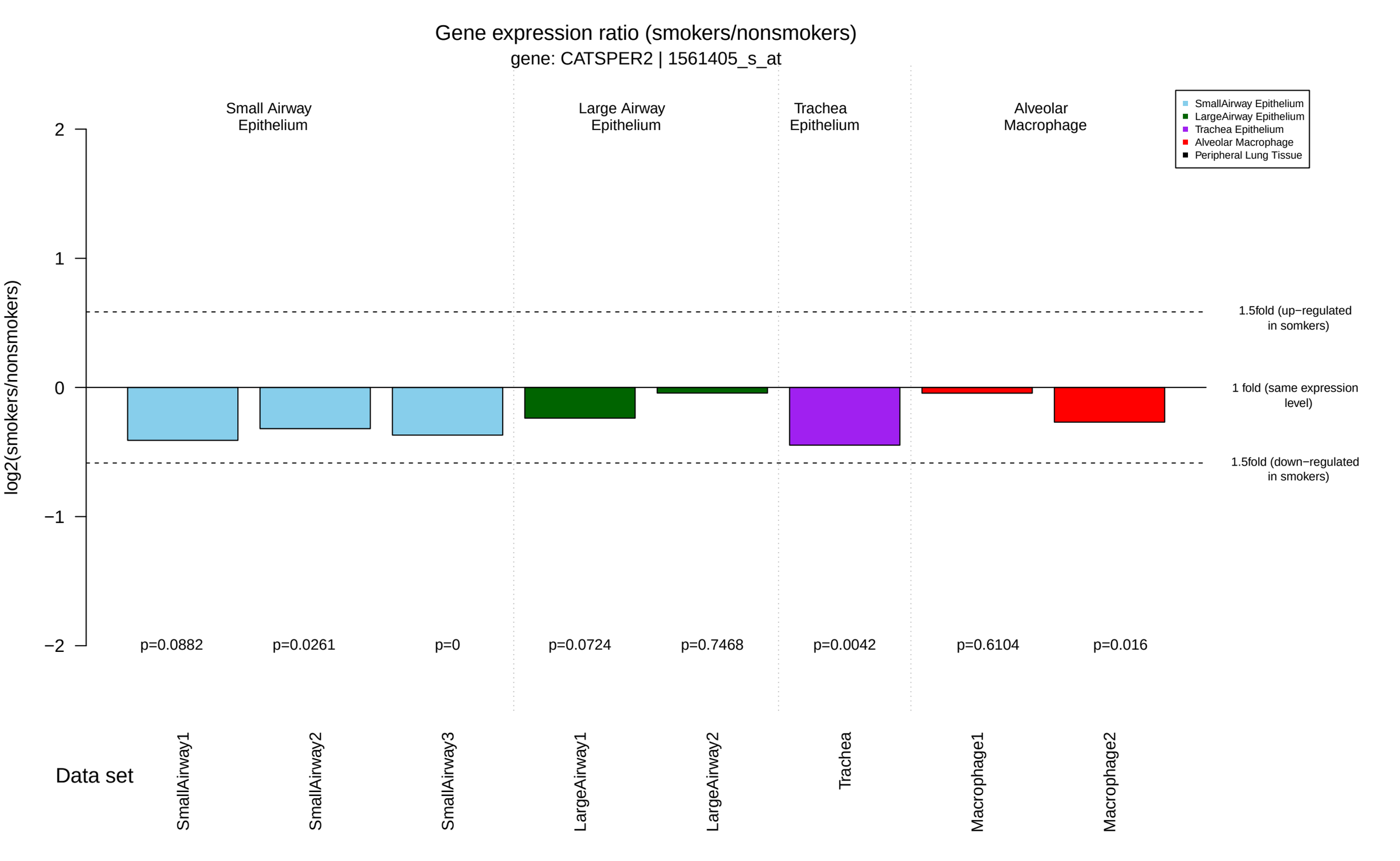
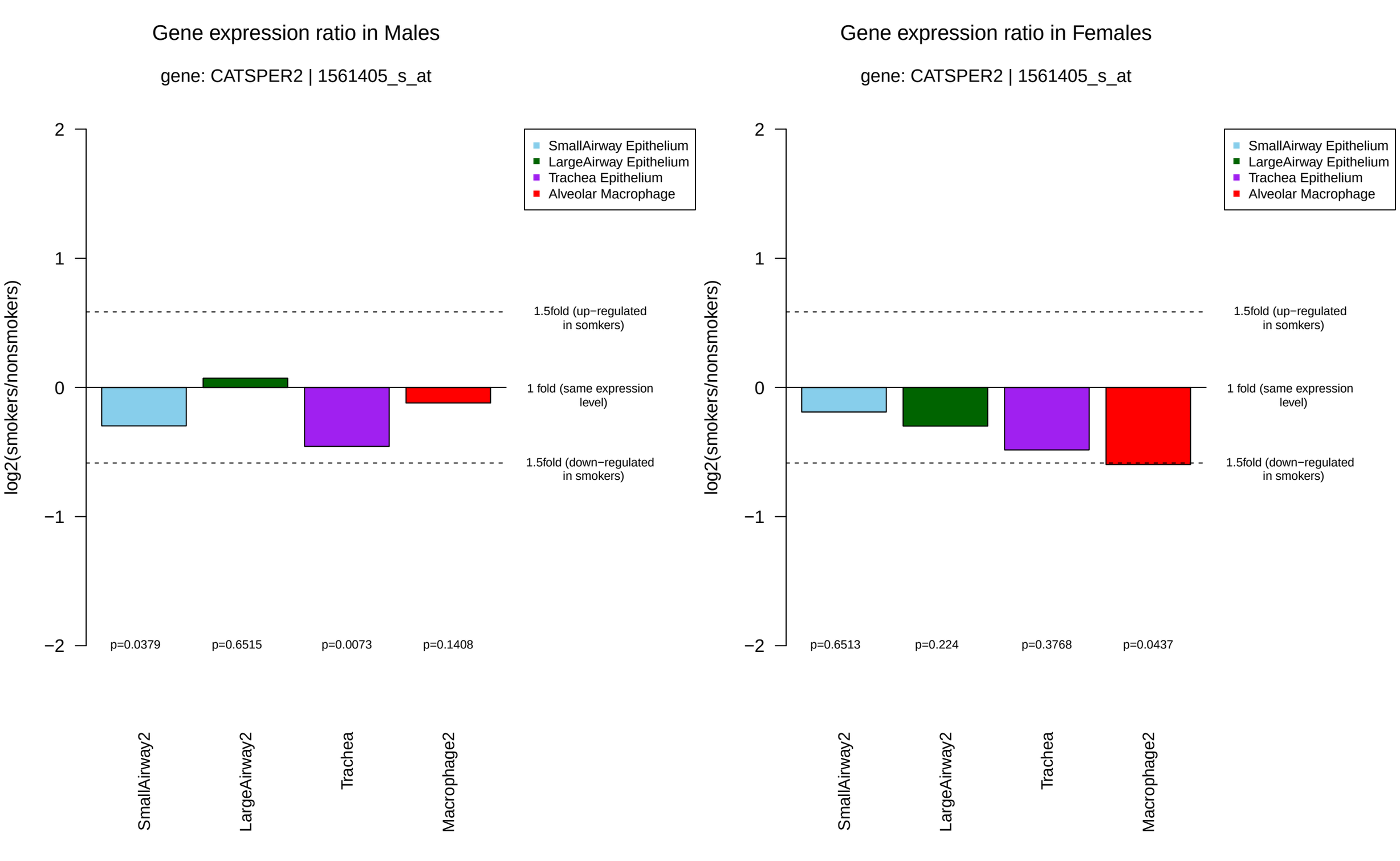
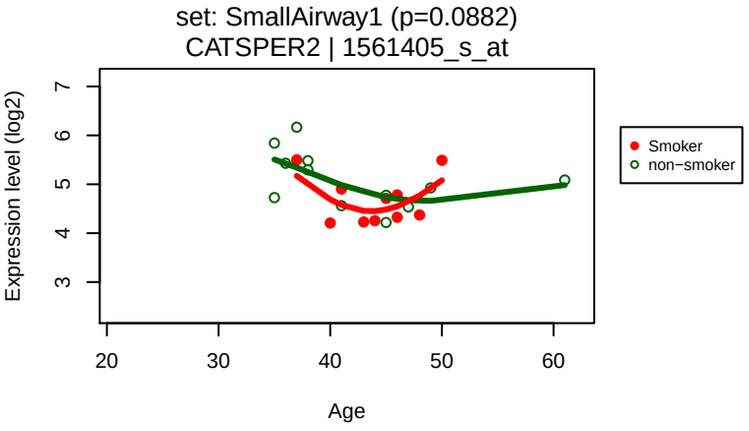
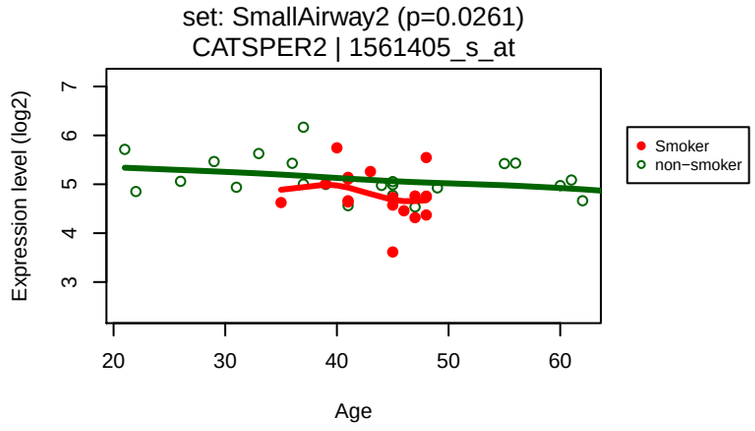
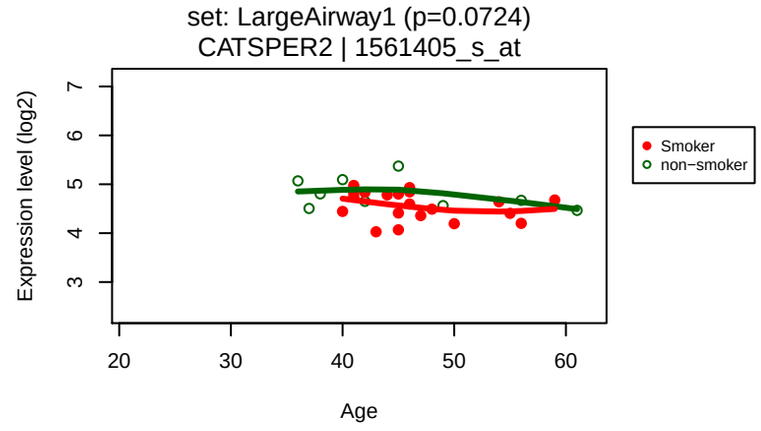

Gene expression of smokers and nonsmokers
gene: CATSPER2 | 1561405_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0882
p=0.0261
p=0
p=0.0724
p=0.7468
p=0.0042
p=0.6104
p=0.016
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

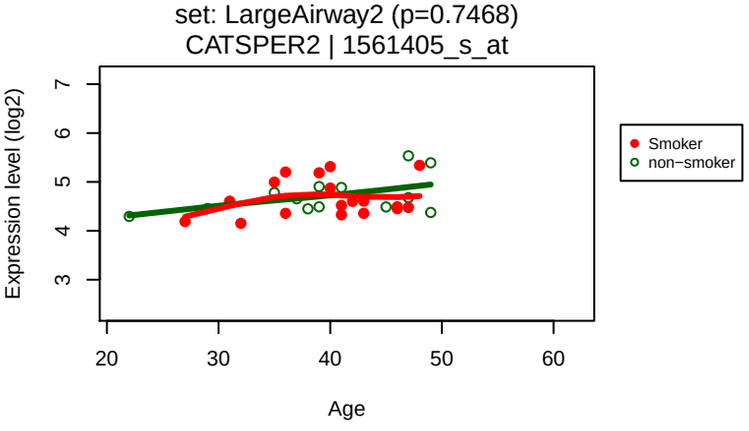

Gene expression ratio (smokers/nonsmokers)
gene: CATSPER2 | 1561405_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0882
p=0.0261
p=0
p=0.0724
p=0.7468
p=0.0042
p=0.6104
p=0.016
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

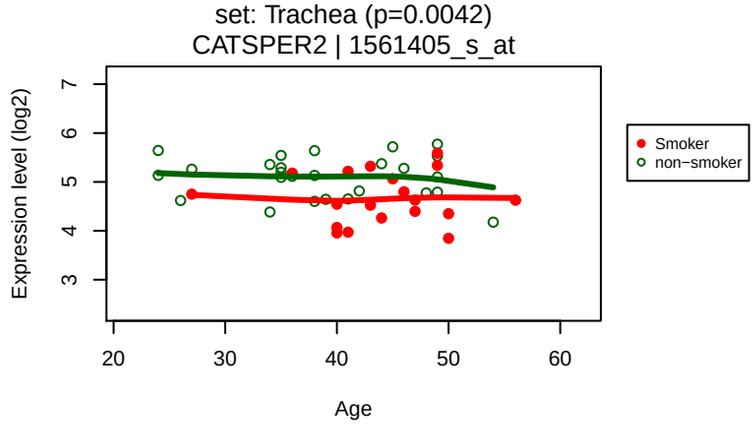

Gene expression ratio in Males
gene: CATSPER2 | 1561405_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0379
p=0.6515
p=0.0073
p=0.1408
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CATSPER2 | 1561405_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6513
p=0.224
p=0.3768
p=0.0437
SmallAirway2
LargeAirway2
Trachea
Macrophage2

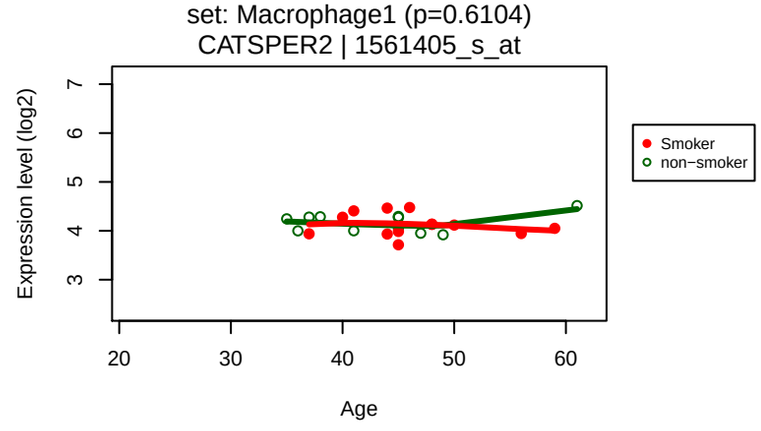

set: SmallAirway1 (p=0.0882)
CATSPER2 | 1561405_s_at
Expression level (log2)
Age
Smoker
non−smoker

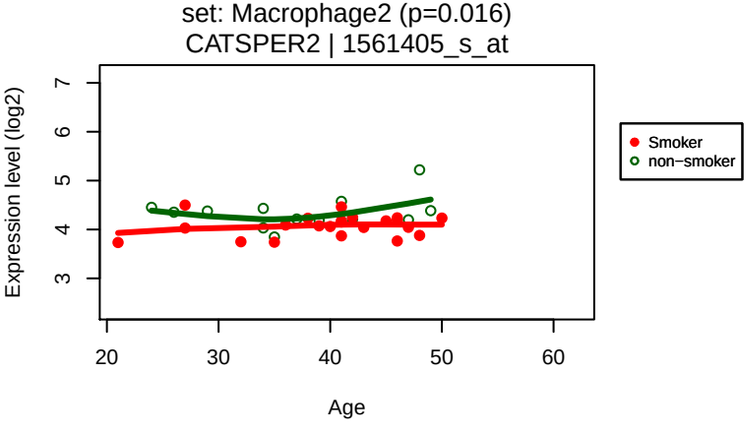

set: SmallAirway2 (p=0.0261)
CATSPER2 | 1561405_s_at
Expression level (log2)
Age
Smoker
non−smoker

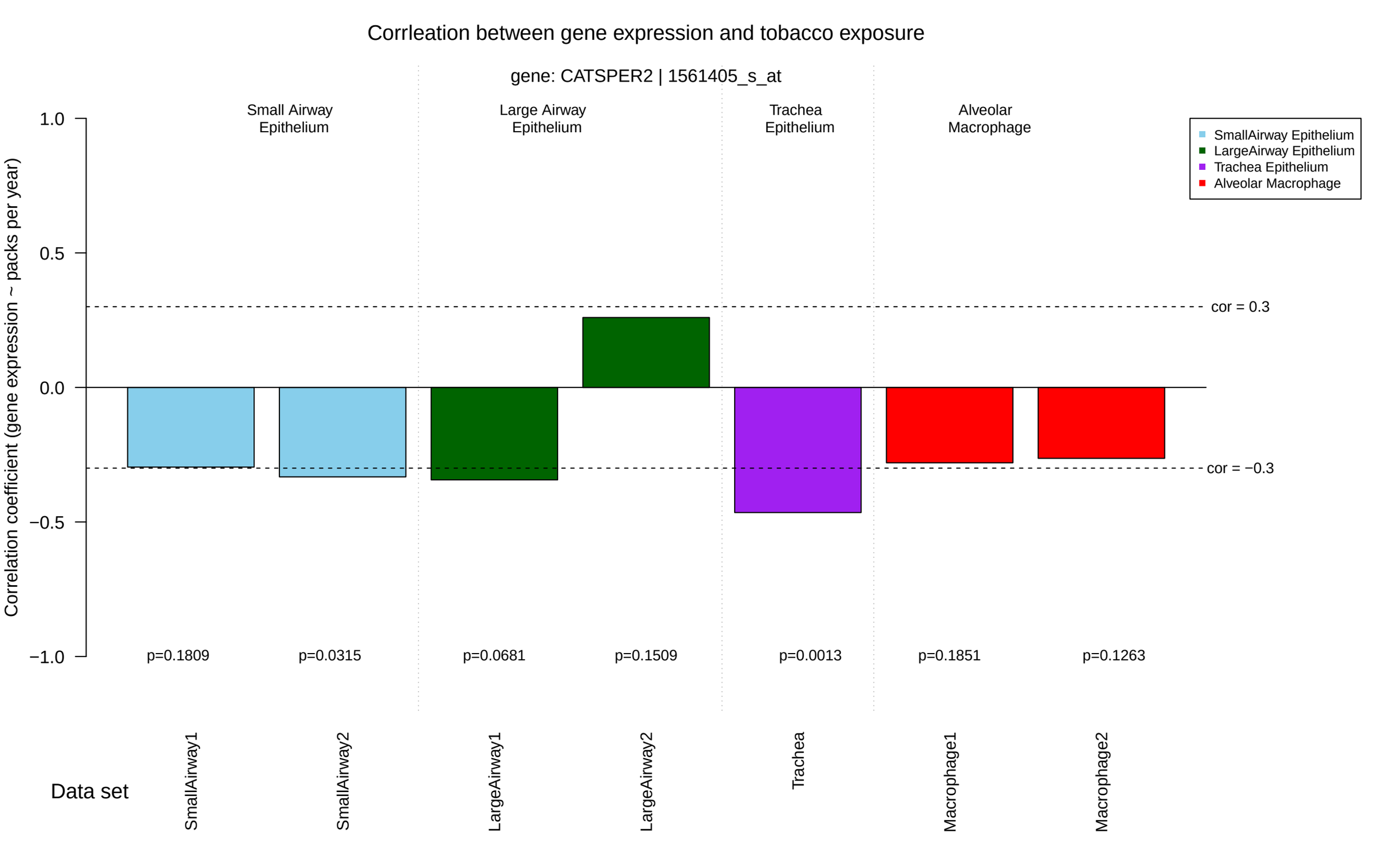

set: LargeAirway1 (p=0.0724)
CATSPER2 | 1561405_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7468)
CATSPER2 | 1561405_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0042)
CATSPER2 | 1561405_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6104)
CATSPER2 | 1561405_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.016)
CATSPER2 | 1561405_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CATSPER2 | 1561405_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1809
p=0.0315
p=0.0681
p=0.1509
p=0.0013
p=0.1851
p=0.1263
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2